

YASSINE BEL HADJ ALI

Electrical & Automation Engineering Student | Embedded Systems, IoT & AI

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PROFESSIONAL SUMMARY

First-year Electrical and Automation Engineering student at École Nationale d'Ingénieurs de Gabès (ENIG) with hands-on experience in embedded systems, IoT architecture, automation, and applied AI. Proficient in C/C++, Python, ESP32 microcontrollers, MQTT messaging, FastAPI backend integration, and AI-assisted workflow automation. Proven track record in developing functional engineering prototypes, managing operations, and leading student initiatives.

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++
- **Embedded Systems & IoT:** ESP32, Arduino, Microcontroller Architecture, Sensors, MQTT, Mosquitto
- **Automation & Web Backend:** FastAPI, REST APIs, Web Development, Dashboards, Git, GitHub
- **AI & Workflow Automation:** Generative AI, AI-assisted Development, Power Automate, Zapier, Claude Code
- **Engineering & Systems:** Electrical Systems, Electronics, Automation & Control, Industrial Systems, Robotics
- **Methodologies & Tools:** System Simulation, Hardware Prototyping, Anomaly Detection, Technical Documentation

ENGINEERING & AI PROJECTS

EnerGuard – Intelligent Industrial Energy Monitoring System

2026

IoT, Embedded Systems, FastAPI, MQTT

ENIG

- Architected an ESP32-based IoT hardware solution to monitor industrial machinery electrical metrics, analyzing power, energy consumption, and power-factor efficiency.
- Developed an end-to-end data pipeline routing real-time sensor data via MQTT/Mosquitto protocols to a FastAPI backend and interactive web dashboard.
- Implemented machine-level energy analytics and anomaly detection mechanisms to optimize industrial energy usage.

FloodGuard AI – Flood-Impact Simulation & Evacuation Support

2026

AI, Python, GIS, Route Optimization

ENIG

- Built a decision-support prototype combining flood scenario modeling, road network GIS data, and real-time route recalculation.
- Integrated an AI copilot agent to deliver evidence-grounded emergency evacuation explanations and automated tool orchestration.
- Processed real-world spatial datasets and open weather APIs while maintaining strict separation between deterministic routing and simulated hazard models.

WORK EXPERIENCE

Founder

2026 – Present

IntellDev

Tunisia

- Founded and managed a digital solutions initiative specializing in web development, UI/UX design, and digital marketing workflows.
- Oversaw technical delivery and client requirements execution across full-stack development projects.

Engineering Intern

2026

Sotualco

Tunisia

- Analyzed automated production lines, control systems, power distribution units, and heavy electrical machinery.
- Investigated the practical integration between electrical, mechanical, and industrial automation components.

Industrial Observer

2026

Chimie Couleur

Tunisia

- Evaluated automated manufacturing processes, material flow, and supply chain logistics within an industrial plant.
- Assessed industrial safety standards, workplace access equipment, and machinery operational procedures.

Customer Service Advisor

Concentrix (Petro-Canada Account)

Aug 2024 – Jan 2025

Tunisia

- Resolved complex customer requests and service queries following strict operational and quality standards.
- Utilized SugarCRM, Microsoft Power BI, and Office Suite to maintain accurate data reporting and performance metrics.

EDUCATION

National Engineering Diploma in Electrical & Automation Engineering

2025 – Present

École Nationale d'Ingénieurs de Gabès (ENIG)

Expected Graduation: June 2029

- Core Coursework: Electrical Systems, Electronics, Automation & Control, Industrial Systems, Signal Processing, Numerical Methods, Embedded Programming.

Preparatory Engineering Cycle – Mathematics & Physics (MP)

2023 – 2025

Institut Préparatoire aux Études d'Ingénieurs de Gabès (IPEIG)

Gabès, Tunisia

CERTIFICATIONS & LEADERSHIP

- Certification – Generative AI & Workflow Automation (GoMyCode, 2026): Applied Generative AI, Power Automate, Zapier, and Claude Code for automated software development workflows.
- Founder & Coordinator – Casa della Musica, ENIG: Created and coordinated a university music club, organizing campus events and supporting student performances.
- Head of Internal Affairs – Interact Tunis Paradise: Managed internal operations for youth-led community service, social impact, and charity projects.
- Active Member – ENIG Robotics Club: Participated in technical robotics initiatives and technical workshops.

LANGUAGES

- **Arabic:** Native · **English:** Professional Working Proficiency (B2) · **French:** Professional Working Proficiency (B2)